to produce 150 tons of copper and 2825 Kilograms of silver. Do not solve the equation.

$\boxed{\text{Soln:}}$ $x_1 v_1 + x_2 v_2 = \begin{bmatrix} 150 \\ 2825 \end{bmatrix}$

29) Let $v_1, \ldots, v_k$ be points in $\mathbb{R}^3$ and suppose that for $j = 1, \ldots, k$ an object with mass m_j is located at point v_j. Physicists call such objects $\boxed{\text{point masses}}$. The total mass of the system of point masses is

$\boxed{m = m_1 + \ldots\ldots + m_k}$.

The center of gravity (or center of mass) of the system is

$$\boxed{\bar{v} = \frac{1}{m}\left[m_1 v_1 + \ldots\ldots + m_k v_k\right]}$$

Compute the center of gravity of the system consisting of the following point masses:

Point	Mass
$v_1 = (5, -4, 3)$	2g
$v_2 = (4, 3, -2)$	5g
$v_3 = (-4, -3, -1)$	2g
$v_4 = (-9, 8, 6)$	1g

$m = (2+5+2+1) = 10g$ $\qquad \bar{v} = \frac{1}{10}\left[2(5,-4,3) + 5(4,3,-2) + 2(-4,-3,-1) + 1(-9,8,6)\right]$

$\bar{v} = \frac{1}{10}\left[(2\cdot 5, 2\cdot(-4), 2\cdot 3) + (5\cdot 4, 5\cdot 3, 5\cdot(-2)) + (2\cdot(-4), 2\cdot(-3), 2\cdot(-1)) + (1\cdot(-9), 1\cdot 8, 1\cdot 6)\right]$ (By Scalar multiplication)

$= \frac{1}{10}\left[(10,-8,6) + (20,15,-10) + (-8,-6,-2) + (-9,8,6)\right]$

$= \frac{1}{10}\left[(10+20+(-8)+(-9)), -8+15+(-6)+8, 6+(-10)+(-2)+6)\right]$

$= \frac{1}{10}\left[(13, 9, 0)\right] = (1.3, 0.9, 0)$